

PRIYANKA ASTHANA

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Research Summary

BTech CSE graduate with a granted Indian patent, two Springer Nature submissions under peer review, and an ACL submission on LLM failure modes in high-stakes domains. Research centers on the reliability of retrieval and generation systems: where embedding geometry does and does not predict downstream model behavior, and how constrained multi-agent architectures perform in medical and cultural-heritage settings. ERA:AI Fellowship Finalist, Technical Track, Cambridge University.

Education

Dr. APJ Abdul Kalam Technical University

2022 – 2026

BTech (Hons.) Computer Science and Engineering, Minor: Robotics

- CGPA: 8.63/10, I-DIV with Honours, Gold Medalist
- Relevant Coursework: Algorithms, Data Structures, Linear Algebra, Probability & Statistics, Optimization Methods, Discrete Mathematics, Theory of Computation

Research & Intellectual Property

Published Patent: AI-Driven Conversational Medical Voice Assistant ↗ | Application No. 202611012869 Mar 2026

- Designed a multi-agent conversational system routing patient symptom inputs to specialist agents (cardiologist, dermatologist, etc.) using NLP-based intent classification and semantic parsing.
- Stress-tested across 500 simulated patient scenarios, achieving 92% disease detection accuracy.

Contextual Embedding Bias Predicts LLM Near-Synonym Confusion ↗ | ACL submission

2026

- Built NEARSYNONYMBENCH, a benchmark of 120 domain-specific near-synonym pairs across medical, Indian cultural heritage, and emergency blood resource management, annotated with confusion probes and stakes levels.
- Showed isolated embedding cosine similarity does not predict LLM confusion (Pearson $r = -0.005$) across four sentence-transformer architectures and three LLM families, while contextual embedding bias (probe-to-term proximity) is a significant predictor ($\rho = 0.299$, $p < 0.001$), explaining 45.3x more variance.
- Identified terminological evasion, accurate description without term commitment, as an underreported failure mode accounting for 28.3% of open-ended LLM responses.

Embedding Model Selection for Domain-Specific RAG ↗ | SN Computer Science (Springer Nature)

2026

- Conducted a systematic benchmark evaluation of three embedding models (MiniLM, MPNet, multilingual MiniLM) on a domain-specific Indian cultural heritage corpus using RAGAS metrics: faithfulness, answer relevancy, and factual correctness.

Priority-Weighted Spatial Framework for Blood Allocation ↗ | Journal of Computational Social Science

2026

- Developed and published a GeoAI framework combining spatial clustering with time-bound token reservation, designed for real-world emergency resource allocation under scarcity conditions.

Research & Industry Experience

IIT BHU | NLP Engineer Intern

Feb 2026 – Jun 2026

- Built an NLP-driven test automation pipeline for heritage text systems, processing 1,000+ text inputs for semantic accuracy validation, directly supporting digital preservation of historical corpora.
- Diagnosed and resolved 20+ NLP parsing failures in live systems, improving pipeline reliability by 25%; documented root cause analysis and recommended architectural fixes to the senior engineering team.

Bharat Space Education Research Centre | AI Research Intern

Dec 2025 – Feb 2026

- Applied ML algorithms to real-time sensor data fusion across 5 experimental UAS modules in simulated space mission environments, contributing to trajectory prediction and a 20% reduction in subsystem latency.

Hindalco Industries, Aditya Birla Group | Business Analyst Intern

Jul 2025 – Sept 2025

- Built SQL queries against Oracle ERP systems to extract, validate, and transform data; identified process inefficiencies through gap analysis and translated findings into Tableau dashboards for senior leadership.
- Communicated performance metrics and analytical insights directly to senior team members, reducing manual processing time by 25% and improving API query performance by 10%.

Projects

Kashivani ↗ | Domain-Specific RAG Pipeline for Cultural Heritage

Sep 2025 – Mar 2026

- Designed and built an end-to-end RAG pipeline for Varanasi cultural heritage knowledge, covering ingestion, chunking, embedding, FAISS-based retrieval, and LLM-powered generation.
- Conducted rigorous embedding model evaluation using the RAGAS framework (faithfulness, answer relevancy, factual correctness), reducing hallucinations by 40% through systematic model selection.

DiagnoseAI ↗ | Multi-Agent Medical Symptom Assessment System

Apr 2025 – Sep 2025

- Architected a voice-first multi-agent system integrating speech-to-text, NLP-based symptom routing, specialist LLM agents (cardiologist, dermatologist, etc.), and structured report generation.
- Built agent orchestration and routing logic using LangChain, enabling structured handoffs between specialist LLM agents and consistent context management across the conversational pipeline.
- Achieved 92% disease detection accuracy across 500 simulated patient scenarios; work patented.

RAKT ↗ | Tokenized GeoAI Emergency Blood Allocation Framework

Dec 2024 – May 2025

- Engineered a spatial-temporal allocation system using K-means clustering on GPS coordinates with haversine distance, applied to real inventory data from e-Raktkosh, India’s national blood transfusion portal.
- Implemented a time-bound token reservation mechanism to prevent concurrent allocation conflicts, solving a real coordination problem in emergency logistics.
- Achieved 100% high-priority allocation and a 6.6% distance reduction under simulated scarcity conditions; framework generalizable to enterprise resource allocation problems.

Technical Skills

AI/GenAI & RAG: RAG pipeline design, embedding model evaluation, constrained LLM reasoning, multi-agent architectures, LangChain, agentic AI workflows, prompt engineering, RAGAS framework, FAISS

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Sentence Transformers, HuggingFace

Data & Analytics: Pandas, NumPy, Tableau, Oracle ERP data pipelines, spatial data analysis (K-means, haversine), SQL

Programming: Python (primary), JavaScript, React, TailwindCSS, Git

Cloud & AI Platforms: Microsoft Azure (AI-900, AI-102 certified), Oracle AI Vector Search (certified)

Certifications

Microsoft Azure AI-900, AI Fundamentals

Microsoft Azure AI-102, Designing and Implementing AI Solutions (Azure OpenAI, Azure AI Search, Responsible AI)

Oracle AI Vector Search Professional Certification, Applied AI Systems

Oracle Cloud Infrastructure Certification, Generative AI Professional

Awards & Honours

ERA:AI Fellowship Finalist, Technical Track

Apr 2026

Cambridge University, UK

- Selected through a competitive multi-stage global application for a fully funded AI safety research fellowship at Cambridge; one of the few finalists shortlisted in the technical track from a worldwide applicant pool.

NPTEL STAR Award

Aug 2025

IIT Kanpur

- Awarded to top-performing students nationally across NPTEL courses; recognises exceptional academic performance evaluated by IIT Kanpur faculty.

Leadership

Student Council President, AITM, Varanasi (May 2025 – May 2026)

Patriotic Youth Ambassador, Veterans India (Aug 2025 – Sept 2025)